

DKHILI FARES

Embedded Systems Engineer | STM32 & IoT

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Embedded systems engineer with hands-on STM32 and multi-vendor MCU experience. Benchmarked STM32Cube middleware stacks against 5 competing ecosystems at STMicroelectronics. Autonomously designed AI-powered automation tools to accelerate firmware analysis workflows.

EDUCATION

Graduate Degree in Computer Engineering , Faculty of Sciences of Tunis	2023 – Present
Relevant coursework: Real-Time Systems, Computer Architecture, Embedded Systems, Digital Design, Networks	
Integrated Preparatory Cycle in Mathematics and Computer Science , Faculty of Sciences of Tunis	2021 – 2023

PROFESSIONAL EXPERIENCE

- **STMicroelectronics** Feb 2026 – Present
Tunis, Tunisia
End-of-Study Internship (PFE) — STM32Cube Ecosystem Benchmark
 - Benchmarked STM32Cube middleware stacks (**USB, FreeRTOS, FatFS**) against **5 competing ecosystems** (TI, NXP, Renesas, GigaDevice) measuring performance and memory footprint
 - Autonomously architected an **AI-powered footprint analyzer** with chatbot interface — automates root-cause gap analysis, scoring, and optimization recommendations across MCU firmware builds
 - Engineered a **custom VS Code Copilot agent** that automates project creation, cross-vendor porting, build configuration, and debug setup, accelerating the benchmark workflow
 - Presented competitive analysis findings to the **STMicroelectronics marketing team**, identifying per-vendor strengths and actionable improvement areas
 - Both tools built on **own initiative** without supervisor request, demonstrating engineering autonomy
- **STMicroelectronics** July – August 2025
Tunis, Tunisia
Summer Internship — Porting EEMBC CoreMark-PRO to STM32
 - Solved a **POSIX-dependency barrier** by porting the CoreMark-PRO benchmark from Linux to bare-metal STM32, adapting the MITH abstraction layers for ARM Cortex-M
 - Executed and compared performance across **Cortex-M4 and Cortex-M7** on 3 STM32 boards (H747I-EVAL, Nucleo-144, Discovery U5)
 - Optimized memory-constrained workloads by leveraging **external SDRAM** and tuning IAR/STM32CubeMX build configurations
- **Vicom Consulting** August 2024
Tunis, Tunisia
Summer Internship — Embedded Linux Development
 - Cross-compiled a **Qt GUI application** for Raspberry Pi using a custom ARM toolchain and deployed it on embedded Linux
 - Configured the build system for **cross-platform compatibility** and validated on-target execution over SSH

PROJECTS

- **AI-Powered Firmware Footprint Analyzer** 2026
Tools: Python, LLM/AI Integration, Chatbot UI, Automated Report Generation
 - Architected a tool that **automates MCU firmware footprint analysis** — classifies binary sections, identifies root causes of size gaps across vendors, and generates optimization recommendations
 - Integrated an **AI chatbot interface** for interactive querying of footprint data and cross-vendor comparison results
- **Benchmark Copilot Agent for STM32** 2026
Tools: VS Code Agent API, TypeScript, STM32CubeMX, Multi-vendor Toolchains
 - Built a **custom VS Code Copilot agent** that automates end-to-end benchmark project scaffolding: creation, cross-vendor porting, build configuration, and debug setup
- **SecureAccess — Multi-Factor Authentication System** May 2025
Tools: Python, Raspberry Pi, OpenCV, Flask, FaceRecognition, RFID
 - Engineered a physical security system combining **3 authentication factors** (RFID, facial recognition, PIN) with a Flask web dashboard for access monitoring
- **Embedded Linux Cross-Compilation** November 2025
Tools: Buildroot, GCC, U-Boot, Raspberry Pi, BeagleBone Black
 - Configured a **cross-compilation toolchain** and customized a minimal embedded Linux image with Buildroot targeting ARM platforms

- **GPRS Communication Library for IoT**
Tools: STM32, C, SIM808, UART
 - Implemented a reusable **STM32 HAL library** for the SIM808 module, abstracting AT commands into clean HTTP GET/POST APIs over GPRS
 - **Network Administration Infrastructure**
Tools: EVE-NG, Cisco IOL, Zabbix, Akvorado, Wireshark
 - Deployed a **multi-VLAN enterprise topology** on EVE-NG with centralized monitoring (Zabbix) and traffic analytics (Akvorado)
 - **IoT Gas and Fire Detection System**
Tools: ESP8266, C++, DHT11, MQ2, MQTT
 - Built an **IoT sensor node** for real-time gas/smoke detection with immediate alert notifications via cloud dashboard

December 2024



March 2025



May 2024



SKILLS

Programming: C, C++ (OOP), Embedded C, Python, VHDL, Bash, Java, TypeScript
Embedded & RTOS: STM32 (CubeMX, CubeIDE, Keil, IAR), ESP32/8266, Raspberry Pi, FreeRTOS, Bare-metal, ARM Cortex-M, Peripheral Drivers (GPIO, UART, SPI, I2C, CAN)
Linux & Build Systems: Buildroot, Yocto, Cross-compilation, Makefile/CMake, Shell scripting, U-Boot
IoT & Connectivity: MQTT, LoRa/WAN, GPRS, Node-RED, Cloud dashboards, Power BI
AI & Automation: LLM integration, AI-powered tooling, OpenCV, VS Code Agent development
FPGA & Digital Design: VHDL, Quartus Prime, ModelSim, FSM design, HW/SW Co-design
DevOps & Tools: Git/GitHub, Docker, Jenkins, SonarQube, CI/CD pipelines
Networks & Security: VLAN, Routing, ACL, Cisco CLI (CCNA), Wireshark, Zabbix, TLS/SSL, VPN
Spoken Languages: French (B2), English (B1), Arabic (Native)

CERTIFICATIONS

- **CCNA: Introduction to Networks (Cisco)**
 - **Fundamentals of Deep Learning (NVIDIA)**

October 2024

April 2025

AWARDS AND DISTINCTIONS

- **Competition Performance**
CTF 101, Securinet FST
 - **3rd place** at CTF 101 (Securinet FST) — solved cryptography and forensics challenges under time constraints

April 2025



PROFESSIONAL AFFILIATIONS

- **Securinets FST Tunis**
 - **ATIA FST: AI Club**
 - **GDG on Campus: FST**

November 2024 - May 2025

2025 - Present

September 2023 - Present